

LAB EXPERIMENT RECORD

Set 12: Website Traffic Analysis

Field	Details
Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	U.Roshitha
Register Number	192425184
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Day	Tuesday

Aim

To analyse website traffic data using a line chart, a bar chart and a data table in R, and to build an interactive dashboard that combines them.

Dataset: Website Traffic

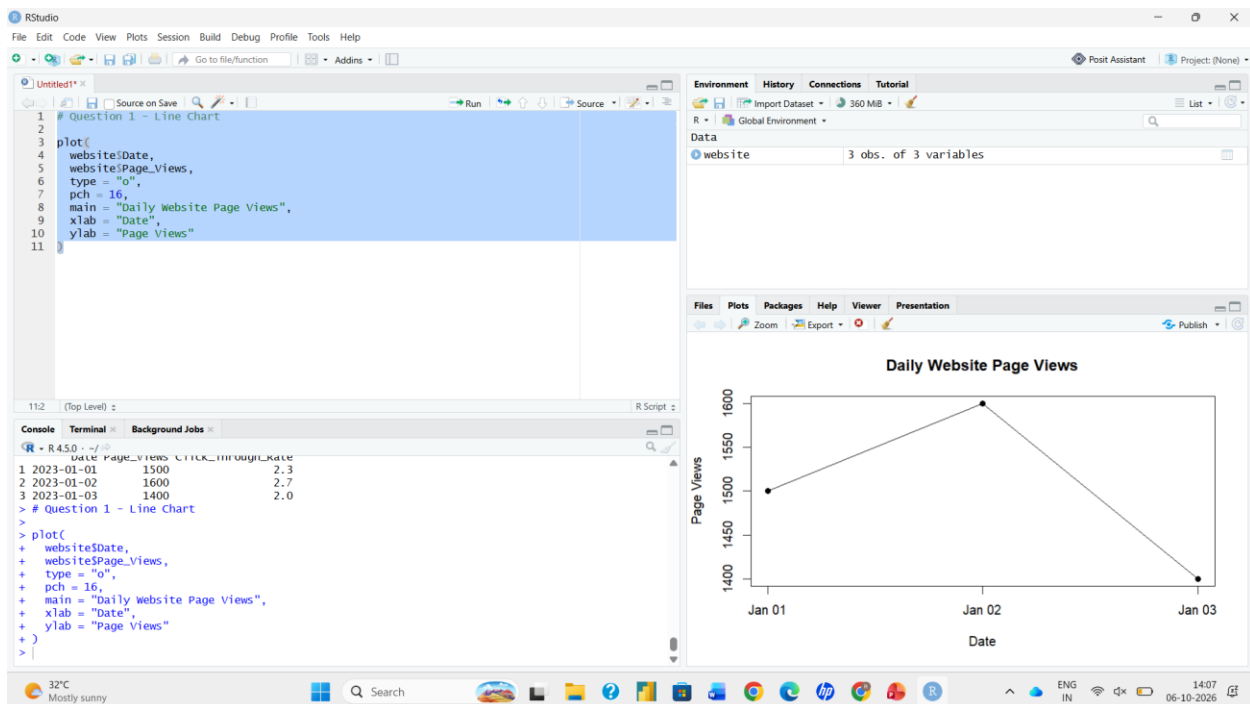
Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  PageViews = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

# Line Chart
plot(traffic$Date, traffic$PageViews,
     type = "o",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views Over Time",
     col = "blue")
```



Output 1: Daily Page Views Line Chart

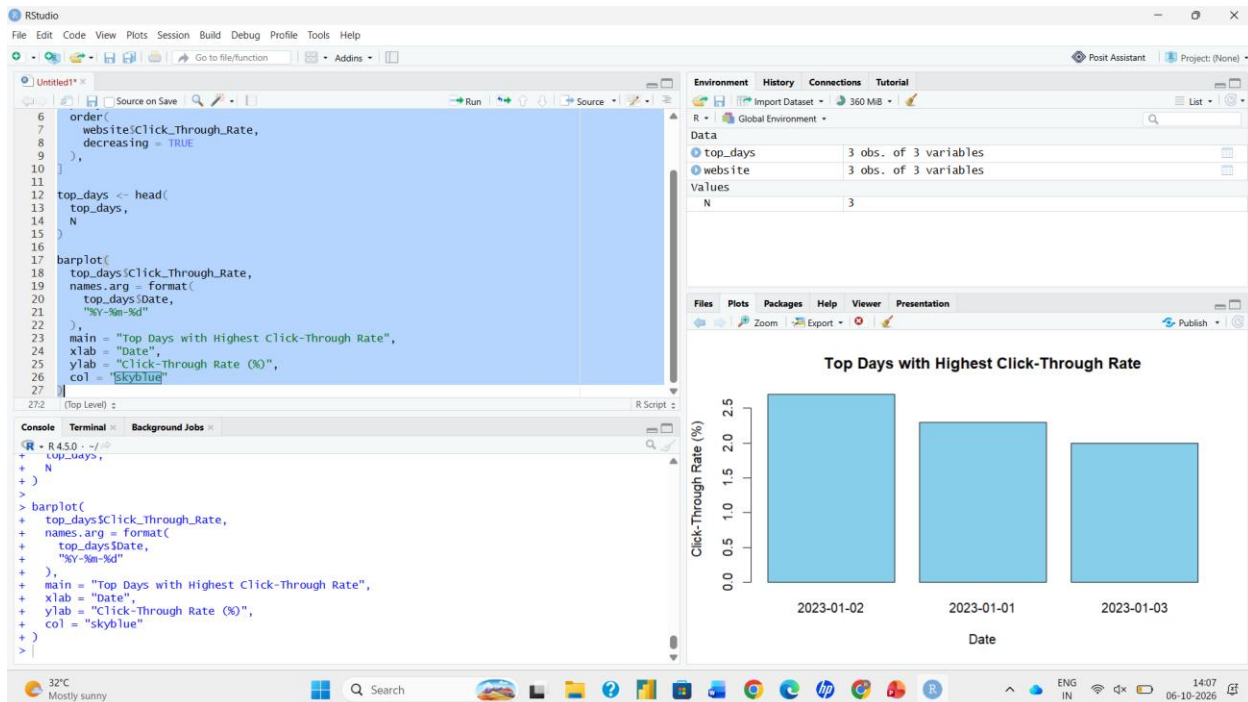
PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Choose N
N <- 3

# Select the top N days by CTR
top_days <- head(traffic[order(-traffic$CTR), ], N)

# Bar Chart
barplot(top_days$CTR,
        names.arg = format(top_days$Date),
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = paste("Top", N, "Days by Click-through Rate"),
        col = "orange")
```



Output 2: Top N Days by Click-through Rate

PROGRAM 3: Website Traffic Table

Build a table to show the website traffic data for each day. Label the table elements.

```
colnames(traffic) <- c("Date", "Page Views", "Click-through Rate (%)")
print(traffic)
View(traffic)
```

The screenshot shows the RStudio environment. The main window displays a data table with 3 columns: Date, Page Views, and Click_Through_Rate. The table contains 3 rows of data. The right sidebar shows the 'Environment' pane with 'traffic' as a data frame. The 'Columns' pane shows the details for the 3 columns: Date (date), Page Views (dbl), and Click_Through_Rate (dbl).

	Date	Page Views	Click_Through_Rate
1	2023-01-01	1500	2.3
2	2023-01-02	1600	2.7
3	2023-01-03	1400	2.0

Showing 1 to 3 of 3 entries, 3 total columns

Output 3: Website Traffic Table

PROGRAM 4: Interactive Shiny Dashboard

Build an interactive dashboard combining the line chart, bar chart and the table using the Shiny package.

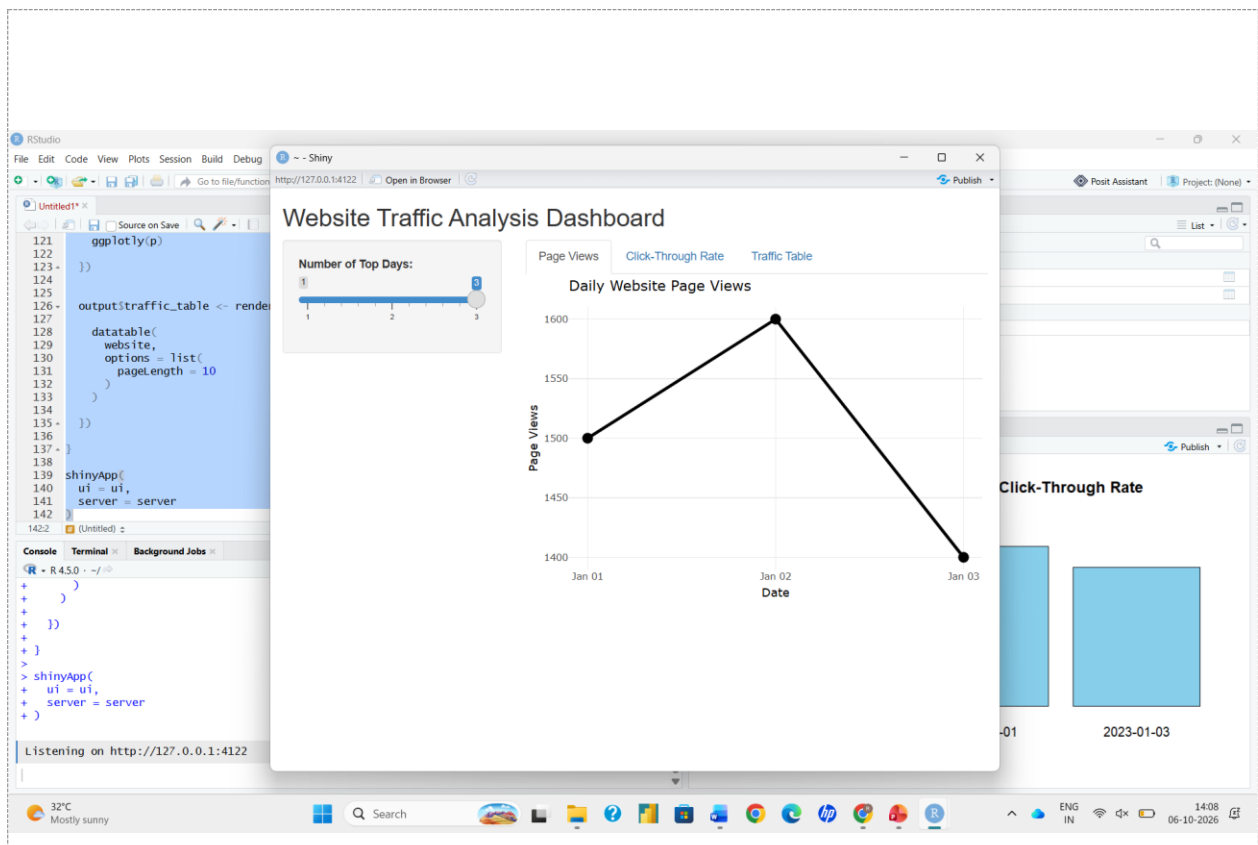
```
library(shiny)

traffic <- data.frame(
  Date    = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  PageViews = c(1500, 1600, 1400),
  CTR      = c(2.3, 2.7, 2.0)
)

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  sidebarLayout(
    sidebarPanel(
      sliderInput("N", "Top N days by Click-through Rate:",
        min = 1, max = nrow(traffic), value = 3, step = 1)
    ),
    mainPanel(
      plotOutput("line"),
      plotOutput("bar"),
      tableOutput("tbl")
    )
  )
)

server <- function(input, output) {
  output$line <- renderPlot({
    plot(traffic$Date, traffic$PageViews, type = "o",
      xlab = "Date", ylab = "Page Views",
      main = "Daily Page Views Over Time", col = "blue")
  })
  output$bar <- renderPlot({
    top_days <- head(traffic[order(-traffic$CTR), ], input$N)
    barplot(top_days$CTR, names.arg = format(top_days$Date),
      xlab = "Date", ylab = "Click-through Rate (%)",
      main = paste("Top", input$N, "Days by Click-through Rate"),
      col = "orange")
  })
  output$tbl <- renderTable({
    d <- traffic
    d$Date <- format(d$Date)
    colnames(d) <- c("Date", "Page Views", "Click-through Rate (%)")
    d
  })
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Dashboard